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From Farm to Forecast: An Integrated Blockchain Framework for Multi-Stakeholder Agricultural Intelligence

System Architecture and Technical Foundations

The development of a comprehensive blockchain-enabled software system for an end-to-end data pipeline necessitates a robust and multi-layered architectural framework. This architecture must transcend the simplistic view of a blockchain as a mere database and instead position it as a foundational layer for establishing trust, ensuring data provenance, and enabling automated execution [91] [211]. The core value proposition of such a system lies in its ability to create a tamper-proof, transparent, and verifiable record of data and transactions across a complex network of stakeholders, thereby addressing fundamental issues of trust and accountability prevalent in traditional agricultural supply chains [9] [49]. The proposed architecture should be designed as a hybrid model, intelligently partitioning data between on-chain and off-chain storage to balance the inherent trade-offs between cost, scalability, performance, and security. On-chain components would be reserved for immutable records of data provenance, access control policies, transactional events like ownership transfers or payments, and the logic of smart contracts themselves. Off-chain storage, utilizing technologies like decentralized storage systems or big data ecosystems such as Hadoop, would house large volumes of raw sensor data, images, and other non-critical information [6] [75]. This strategy directly addresses the significant challenge of blockchain scalability, where storing vast datasets on-chain is computationally expensive and inefficient [8]. By recording only metadata, cryptographic hashes, or pointers to off-chain data on the blockchain, the system ensures that the ledger remains lightweight and responsive while still preserving the integrity and traceability of the underlying assets.

The selection of a consensus mechanism is a critical technical decision that profoundly impacts the system's performance, security, and suitability for the intended application. For an agricultural supply chain context involving numerous participants, energy-intensive Proof-of-Work (PoW) mechanisms are ill-suited due to their high computational overhead and slow transaction finality [9]. Instead, permissioned or consortium blockchain models employing more efficient consensus algorithms are strongly

recommended. Mechanisms such as Practical Byzantine Fault Tolerance (PBFT), Delegated Proof of Stake (DPoS), and Proof of Authority (PoA) offer superior throughput and lower latency, making them viable for resource-constrained settings and real-time applications [9] [69]. These permissioned systems allow for a curated set of trusted nodes to validate transactions, which enhances scalability and reduces operational costs compared to public, permissionless networks. The choice of a specific blockchain platform, such as Hyperledger Fabric or Ethereum, would depend on the desired level of decentralization, privacy requirements, and integration capabilities [7]. For instance, Hyperledger Fabric provides a modular architecture that supports pluggable consensus and a sophisticated identity management system, making it well-suited for enterprise-grade applications within a consortium of known entities [6].

To ensure the system is technically feasible and performs reliably under real-world conditions, performance benchmarking and optimization strategies must be integral to the design process. A prototype study integrating Hyperledger Fabric with a Hadoop Big Data ecosystem demonstrated promising results, achieving a throughput of approximately 600 transactions per minute with an average response time of 500 milliseconds under normal load [6]. However, the same study identified a significant bottleneck, noting that maximum transaction latency increased sharply when the sending rate approached 200 Transactions-per-Second (TPS), primarily during the validation phase [6]. This highlights the necessity of proactive optimization. Potential strategies include deploying additional endorsing peers or validators to distribute the processing load, dynamically adjusting block sizes based on transaction rates, or implementing sharding techniques to partition the ledger and parallelize validation processes [6]. Furthermore, the system must be architected to handle intermittent connectivity, a common issue in rural agricultural environments. Infrastructure-less networking concepts, which connect devices end-to-end without relying on fixed facilities, could be leveraged to maintain local continuity during brief outages [15][21][73]. The overall technical framework must also incorporate robust identity management and access control mechanisms from the outset. This involves integrating standards like XACML for attribute-based access control and leveraging Self-Sovereign Identity (SSI) principles through Decentralized Identifiers (DIDs) and Verifiable Credentials (VCs) to give participants granular control over their data and claims [40] [41]. This layered approach, combining a suitable consensus mechanism, a hybrid data storage model, performance optimization planning, and integrated identity management, forms the technical bedrock upon which a scalable, secure, and trustworthy data-driven ecosystem can be built.

Component	Description	Rationale
Consensus Mechanism	Practical Byzantine Fault Tolerance (PBFT), Delegated Proof of Stake (DPoS), or Proof of Authority (PoA).	High efficiency, low latency, and scalability compared to PoW; suitable for consortium models [9] [69].
On-Chain Storage	Immutable ledger for: data provenance hashes, access control policy commitments, smart contract code, and key transactional events.	Guarantees data integrity, transparency, and auditability for critical records [6] [49].
Off-Chain Storage	Utilizes decentralized storage (e.g., IPFS) or big data platforms (e.g., Hadoop) for bulk data (sensor readings, images).	Manages system scalability and reduces on-chain costs by avoiding direct storage of large datasets [6] [75].
Identity Management	Integration of Self-Sovereign Identity (SSI) using DIDs and Verifiable Credentials (VCs) [41] [42].	Provides users with control over their digital identities and enables selective disclosure of attributes [39].
Access Control	Policy-based access using standards like XACML, enforced by smart contracts.	Enables fine-grained, dynamic control over data sharing, allowing owners to define precise rules for access [38] [40].
Performance Optimization	Strategies include deploying more endorsers/peers, adjusting block size, and implementing sharding.	Proactively addresses potential bottlenecks in transaction throughput and latency to ensure real-world viability [6].

Stakeholder-Centric Value Propositions and Mitigation Strategies

A successful blockchain-enabled data system must deliver tangible value to all stakeholders by explicitly addressing their unique pain points at each stage of the pipeline. The framework must be designed not as a monolithic technology, but as a socio-technical ecosystem where the needs of diverse actors—from smallholder farmers to multinational corporations—are balanced and addressed cohesively. The primary stakeholders in an agricultural data pipeline include data providers (farmers), commercial partners (processors, buyers, retailers), research entities (universities, NGOs), and end-users (consumers, financial institutions). Each group enters the system with distinct motivations, resources, and challenges that must be systematically analyzed and resolved. For data providers, particularly small-scale farmers, the principal barriers often involve high upfront costs for digital tools, poor infrastructure limiting connectivity, low levels of digital literacy, and a deep-seated fear of data misuse by powerful intermediaries [77] [84] [165]. To mitigate these issues, the system must prioritize affordability and accessibility. This can be achieved by integrating low-cost IoT devices, such as those based on NB-IoT for smart irrigation, which significantly reduce capital expenditure and power consumption [52] [183]. Furthermore, the system must empower farmers by giving

them sovereign control over their own data, transforming it from a corporate asset into a monetizable resource. This is facilitated through tokenomics, where farmers receive compensation for contributing verified, high-quality data, thereby creating a direct economic incentive for participation [47] [162].

For commercial partners and researchers, the dominant pain points revolve around data fragmentation, high integration costs with disparate legacy systems, and a lack of clarity regarding data ownership rights [14] [56]. These stakeholders require seamless access to aggregated, high-quality, and interoperable data to conduct market analysis, optimize supply chains, and advance scientific research. The proposed system addresses these challenges through a robust interoperability strategy and a standardized data access framework. By employing cross-chain bridges and adhering to established data exchange protocols, the platform can break down data silos and facilitate the flow of information across different blockchain networks and traditional IT systems [12] [96]. To manage access securely and efficiently, the system will implement a combination of Self-Sovereign Identity (SSI) and Attribute-Based Access Control (ABAC) governed by the XACML standard [40] [41]. This allows data owners to define precise, policy-driven permissions, granting commercial partners or researchers access to specific datasets under defined conditions without compromising broader data security [40]. For example, a researcher could be granted access to anonymized soil health data from a specific region for a set period, while a food processor could be authorized to view real-time temperature logs for a particular shipment of produce [39]. This fine-grained control not only enhances security but also builds trust among collaborating entities. The system's architecture, which separates on-chain provenance from off-chain data storage, further aids commercial partners by reducing the complexity and cost of interacting with the blockchain itself, as they can query off-chain databases that are updated based on verified on-chain events [6].

As the data moves through the pipeline, the value proposition evolves. For agronomists and farm managers, the system offers descriptive and diagnostic analytics that provide a holistic view of operations, enabling them to make more informed decisions about inputs, planting schedules, and resource allocation [116] [201]. For financial institutions and insurance companies, the system provides the reliable, auditable data streams necessary to develop innovative products like pay-for-performance contracts or automated crop insurance claims [36] [54]. When a predictive model confirms a weather event that threatens a crop, a smart contract can automatically trigger an insurance payout, drastically reducing administrative overhead and improving farmer resilience [32]. Finally, for end-users, the ultimate beneficiaries, the system enhances consumer trust by providing verifiable proof of a product's origin, quality, and sustainability credentials, combatting counterfeiting

and fraud [26] [35]. This transparency empowers consumers to make more informed purchasing decisions and creates a positive feedback loop, where ethically sourced products can command premium prices, further incentivizing good practices up the supply chain. By systematically mapping these pain points and designing targeted solutions at each stage—from initial data collection to final decision execution—the framework ensures that every participant has a clear and compelling reason to engage, fostering a truly collaborative and value-generating ecosystem.

The Token Economy: Aligning Incentives and Ensuring Sustainability

The long-term viability of a multi-stakeholder blockchain ecosystem hinges on the creation of a robust and well-designed token economy. Tokens serve as the economic engine that aligns the disparate interests of various participant groups, incentivizes desirable behaviors like data contribution and quality assurance, and establishes a sustainable revenue model for the platform's operation [1] [2] [20]. Without such an economy, participation may wane, leading to data scarcity and ultimately the failure of the system. The design of this tokenomic model requires a step-by-step approach, synthesizing contributions from existing literature to create a framework that is both strategic and adaptable [2] [131]. The first step is to define the utility of the native token. Its functions should extend beyond simple payment to encompass governance rights, access to premium services, and staking mechanisms for network security. For instance, token holders could gain voting rights on key platform decisions, such as protocol upgrades or changes to fee structures, thereby fostering a sense of collective ownership and participatory governance [132]. Tokens could also be used to pay for accessing advanced analytics dashboards or premium data sets offered by researchers or commercial partners, creating a direct B2B revenue stream [70]. Furthermore, tokens can be distributed as rewards to stakeholders who contribute valuable data or perform essential network functions, such as validating transactions in a Proof-of-Stake model.

Incentive structures must be carefully calibrated to motivate participation across all tiers of the ecosystem. For data providers, such as farmers, the primary incentive is financial remuneration for their data. The system can be designed to reward farmers based on the volume and quality of the data they provide, with quality being verifiable through the system's inherent data provenance features [162]. Smart contracts can automate these payments, ensuring fair and transparent compensation. This transforms data collection

from a potential burden into a new source of income, directly empowering smallholders [47]. For commercial partners and researchers, incentives might take the form of access tokens, which grant them the right to query specific datasets or use the platform's analytical tools. They could purchase these tokens using fiat currency or stablecoins, with the proceeds funding platform maintenance and development. Another strategy involves creating profit-sharing schemes where commercial partners contribute a percentage of revenues generated from insights derived using the platform's data, with these profits then distributed back to data contributors via the token economy [60]. This creates a virtuous cycle where investment in the platform yields returns that are reinvested into the community, encouraging continued data contribution. The distribution of tokens at launch is another critical consideration. It must be equitable to ensure broad participation and prevent centralization of power. A mix of allocation methods, such as public sale, private sale for strategic partners, and grants for early developers and community builders, can help achieve this balance [48].

Beyond financial incentives, the token economy can be used to foster a culture of quality and trust within the ecosystem. For example, stakeholders who consistently provide accurate and timely data could earn higher reputation scores, which might translate into higher rewards or better access privileges. Conversely, malicious actors who attempt to manipulate data could face penalties, such as slashing of staked tokens or temporary suspension of access rights. This mechanism leverages game theory to encourage honest behavior. The overall economic model must also address business viability. While user-facing services might be free or freemium, the platform needs a sustainable way to cover its operational costs, including transaction fees on the blockchain (like Ethereum gas fees), server maintenance, and development expenses [8]. Revenue can be generated through a combination of service fees, premium subscriptions for advanced analytics, and transaction fees on any marketplace features built into the platform. Analyzing different fee structures, such as constant fees versus revenue-based or profit-based schemes, can help determine the optimal model for maximizing firm value and user adoption simultaneously [60]. Ultimately, a thoughtfully designed token economy acts as the binding agent for the entire ecosystem, translating abstract goals of transparency and collaboration into concrete economic realities that drive participation and ensure the platform's longevity.

Governance, Interoperability, and Security Protocols

Effective governance, seamless interoperability, and robust security are the three pillars supporting a functional and trustworthy blockchain-based data ecosystem. Governance defines the rules of the system, ensuring that all participants have a clear understanding of their rights, responsibilities, and the mechanisms for resolving disputes. Given the multi-stakeholder nature of the agricultural pipeline, a centralized authority is undesirable and impractical. Instead, a decentralized governance model is required. This could manifest as a multi-stakeholder council representing farmers, commercial partners, researchers, and technical experts, or a more formal on-chain governance system where decisions are made through voting by token holders [132]. The latter model, while potentially slower, offers greater transparency and resistance to censorship. Regardless of the structure, the governance framework must establish clear protocols for proposing and implementing changes to the system's rules, smart contract logic, and economic parameters. This includes defining quorum requirements, voting periods, and mechanisms for executing approved changes, ensuring that no single entity can unilaterally alter the system's fundamental properties. The IEEE Std 2418.5™-2025 provides a useful guide for developing reference architectures and interoperability standards within such frameworks, emphasizing the need for structured and compliant development practices [38].

Interoperability is paramount for breaking down the data silos that plague modern supply chains and enabling the free flow of information between the blockchain network and external legacy systems [10] [14]. A system designed in isolation, no matter how advanced, will have limited utility if it cannot communicate with other platforms used by partners, regulators, or financial institutions. Achieving interoperability requires adherence to open standards and the implementation of specialized protocols. Semantic layer interoperability, for instance, involves defining and encoding data using common ontologies and reference architectures, such as ISO/IEC PRF TR 20547-1, to ensure that data elements have consistent meanings across different systems [38] [46]. At the network level, cross-chain communication protocols and interoperability bridges are essential for transferring assets and data between different blockchain networks [12]. The POLARIS architecture, for example, demonstrates a method for enabling cross-domain access control by integrating self-sovereign identity with a blockchain-based registry, showcasing a practical approach to interoperability [39]. The AgroChain architecture similarly presents variants designed for multi-blockchain environments, highlighting the importance of building systems that can operate across multiple networks to ensure redundancy and flexibility [7]. These efforts collectively aim to reduce fragmentation and create a more cohesive digital landscape for agriculture [12].

Security is a non-negotiable requirement, especially given the increasing sophistication of cyber threats targeting decentralized systems and IoT devices [4] [141]. The system must adopt a defense-in-depth strategy, securing not just the blockchain itself but the entire ecosystem, from the edge devices to the user interfaces. The inherent immutability and decentralization of the blockchain provide a strong foundation for data integrity, preventing unauthorized alteration of records [211]. However, the surrounding infrastructure presents vulnerabilities. IoT sensors and actuators, for example, can be physically compromised or hacked, leading to the injection of malicious data into the system [8]. To mitigate this, a robust device authentication and registration process must be implemented before any device can participate in the network [6]. Secure communication channels, such as those using TLS, and mutual authentication protocols should be mandated for all interactions [6]. Smart contracts, which represent the programmable logic of the system, are a frequent target for exploits. Therefore, they must undergo rigorous auditing and testing before deployment. The integration of AI-powered frameworks can even assist in optimizing smart contracts for security and efficiency [208]. Finally, protecting user privacy is a critical aspect of security. Technologies like Verifiable Credentials (VCs) and privacy-preserving protocols allow users to prove claims about themselves without revealing sensitive personal information [39] [45]. By combining a clear governance model, a commitment to interoperability standards, and a comprehensive security posture, the system can build the trust necessary for widespread adoption and long-term success.

A Phased Analytics Roadmap for Actionable Intelligence

The true power of the proposed data pipeline is realized when it transitions from simply storing and verifying data to generating actionable intelligence that drives improved outcomes. This transformation is best achieved through a phased analytics roadmap that progressively builds capabilities from descriptive insights to prescriptive recommendations. This staged approach allows stakeholders to gradually adapt to new ways of working and derive value at each step, rather than attempting to implement a complex, fully-automated system from day one. The four types of analytics—descriptive, diagnostic, predictive, and prescriptive—form the basis of this roadmap [194][195]. Descriptive analytics serves as the foundational layer, focusing on summarizing historical data to answer the question, "What happened?" [196]. For farmers and supply chain managers, this translates into real-time dashboards that visualize key performance indicators such as crop yields, input costs, field-level environmental conditions, and

logistics tracking data [66] [177]. These dashboards provide the visibility needed to understand past performance and identify immediate areas for improvement [114]. The primary challenge at this stage is aggregating data from disparate sources into a coherent and intuitive format. This requires adopting semantic frameworks and data ontologies to ensure data from different sensors and systems can be meaningfully combined and presented [81].

Once a solid descriptive foundation is in place, the system can evolve to include diagnostic analytics, which aims to answer the question, "Why did it happen?" [116]. This involves applying statistical and analytical techniques to drill down into the data and uncover the root causes of observed trends or anomalies. For example, if a dashboard shows a sudden drop in yield in a specific field, diagnostic tools can correlate this with other available data—such as soil moisture levels from IoT sensors, fertilizer application records, and historical weather patterns—to identify the likely cause [116]. This deeper level of insight helps stakeholders move beyond reactive problem-solving to proactive management. The next phase introduces predictive analytics, which uses historical data to estimate future outcomes and answer the question, "What is likely to happen?" [116]. Machine learning models can be trained on the rich, provenanced dataset stored in the system to forecast crop yields, predict market demand fluctuations, anticipate equipment failures, or identify the likelihood of pest infestations or disease outbreaks [78] [87]. The availability of high-quality, granular data from the blockchain-enabled collection and storage layers is crucial for training accurate and reliable predictive models [125]. Existing learning analytics platforms provide valuable lessons on the challenges of effectively integrating and visualizing predictive outputs for end-users [144][145].

The pinnacle of the analytics roadmap is prescriptive analytics, which answers the question, "What should we do about it?" [116]. Building upon the outputs of descriptive, diagnostic, and predictive analytics, prescriptive systems go a step further by recommending specific courses of action and simulating potential outcomes of different decisions [146][149]. For a farmer, a prescriptive dashboard might not only predict a coming drought but also recommend an optimal irrigation schedule, suggest alternative drought-resistant crops, and simulate the financial impact of each option [126]. This level of intelligence transforms the system from a passive information repository into an active decision support tool. The bridge between analytical insights and physical or financial actions is the smart contract. Prescriptive recommendations can trigger automated workflows; for instance, a recommendation to apply a specific treatment could automatically generate a purchase order for the required inputs and schedule a drone application task [53]. However, for high-stakes decisions, a Human-in-the-Loop (HITL)

architecture is essential. This approach combines the analytical power of AI with human expertise and judgment, ensuring that critical decisions are not made in a purely autonomous fashion [27] [28]. The system should present AI-generated recommendations to a human expert, who can review, validate, and approve the action, thereby mitigating risks and building trust in the technology [30] [31]. This phased integration of analytics capabilities, culminating in a robust HITL framework, ensures that the system delivers increasing value over time while remaining grounded in the practical realities of decision-making.

Usability, Adoption, and Risk Mitigation

While the technical architecture, economic model, and analytical capabilities form the core of the proposed system, its ultimate success is contingent upon its usability, the willingness of stakeholders to adopt it, and the effective mitigation of associated risks. Technology alone is insufficient; a system is only as good as its ability to be used effectively by its intended audience [66]. For stakeholders like smallholder farmers, who may have limited digital literacy and access to high-speed internet, a complex or unintuitive interface can be a significant barrier to adoption [77] [84]. Therefore, user-centered design principles must be embedded throughout the development process. This involves employing participatory design methodologies, engaging stakeholders from all groups—including farmers, agronomists, and supply chain managers—in the design and testing phases to ensure the final product meets their actual needs and workflows [176]. Empirical studies on e-commerce platforms for agriculture have demonstrated the effectiveness of user-centered approaches in identifying usability challenges and improving adoption rates [100][102]. The user interface should be clean, visually intuitive, and focus on delivering actionable insights through well-designed dashboards rather than overwhelming users with raw data [67]. Features such as multilingual support, offline functionality for areas with poor connectivity, and low-bandwidth modes are critical for inclusivity and accessibility [7] [214]. Adherence to web accessibility standards, such as WCAG 2.1 Level AA, is also essential to ensure the system is usable by individuals with disabilities, including provisions for screen-reader support and keyboard navigation [220] [221].

Adoption is driven not only by usability but also by demonstrating clear and immediate value. The system must solve real problems and provide benefits that are easily understood by all stakeholders. For farmers, this means showing how the system can lead

to increased yields, reduced input costs, or better market prices [201]. For commercial partners, the value lies in enhanced supply chain efficiency, improved quality control, and access to richer market intelligence. Overcoming the digital divide requires a concerted effort to address the infrastructural and educational gaps that exist in many rural areas [84]. Partnerships with local extension services, mobile network operators, and government agencies can help provide the necessary connectivity, hardware, and training to facilitate widespread adoption [143]. The concept of Digital Public Infrastructure (DPI) highlights the role such platforms can play in accelerating economic growth and building public trust by providing foundational digital services [204].

Finally, a comprehensive risk management strategy is essential to navigate the uncertainties inherent in deploying a novel, interconnected system. Key risks include cybersecurity threats, legal and regulatory ambiguity, and ethical concerns related to AI and data use. Cybersecurity represents a critical challenge, requiring a multi-layered defense strategy that secures the blockchain, IoT devices, backend systems, and user interfaces [4]. Legal and regulatory frameworks governing data ownership, smart contract enforceability, and token-based transactions are still evolving and vary significantly across jurisdictions, posing a major hurdle for global deployment [30]. Proactive engagement with policymakers and the development of clear terms of service are necessary to navigate this complex landscape. Ethical risks, particularly algorithm aversion—the tendency of humans to distrust and reject recommendations from algorithms—must be addressed through the implementation of explainable AI (XAI) techniques and a robust Human-in-the-Loop (HITL) architecture [174][210]. By prioritizing user-centric design, focusing on demonstrable value, and proactively managing technical, legal, and ethical risks, the system can build the trust and confidence needed for successful and sustainable adoption across the entire agricultural ecosystem.

Reference

1. [PDF] Designing a Token Economy: Incentives, Governance ... - arXiv.org https://arxiv.org/pdf/2602.09608
2. (PDF) Designing a Token Economy: Incentives, Governance, and ... https://www.researchgate.net/publication/390271393_Designing_a_Token_Economy_Incentives_Governance_and_Tokenomics

3. Tokenizing circularity in agri-food systems: A conceptual framework ... https://www.sciencedirect.com/science/article/pii/S0959652623016852
4. Towards Empowering Stakeholders Through Decentralized Trust ... https://www.mdpi.com/2410-387X/9/3/52
5. Understanding token-based ecosystems – a taxonomy of blockchain ... https://link.springer.com/article/10.1007/s12525-020-00396-6
6. IoT Big Data provenance scheme using blockchain on Hadoop ... https://link.springer.com/article/10.1186/s40537-021-00505-y
7. (PDF) Smart Contracts for Managing the Agricultural Supply Chain https://www.researchgate.net/publication/382922207_Smart_Contracts_for_Managing_the_Agricultural_Supply_Chain_A_Practical_Case_Study
8. Blockchain and smart contract for IoT enabled smart agriculture - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC8022535/
9. A Survey of Blockchain Applications for Management in Agriculture ... https://www.mdpi.com/1999-5903/17/1/40
10. [PDF] Exploring Blockchain Interoperability: Frameworks, Use Cases, and ... https://arxiv.org/pdf/2601.02949
11. Blockchain Interoperability in Data Exchange Logistics Integration https://www.mdpi.com/2227-7390/12/10/1516
12. Blockchain Interoperability: Performance and Security Trade-Offs https://dl.acm.org/doi/10.1145/3560905.3568176
13. Blockchain Meets Sharing Economy: A Case of Smart Contract ... https://www.researchgate.net/publication/364707250_Blockchain_Meets_Sharing_Economy_A_Case_of_Smart_Contract_Enabled_Ridesharing_Service
14. [PDF] Blockchain Interoperability - World Bank Documents and Reports https://documents1.worldbank.org/curated/en/373781615365676101/pdf/Blockchain-Interoperability.pdf
15. Enhancing blockchain interoperability and intraoperability ... https://www.tandfonline.com/doi/full/10.1080/17517575.2023.2296647
16. A Review on Blockchain-Based Trust and Reputation Schemes in ... https://www.mdpi.com/2410-387X/9/4/74
17. SoK: Measuring Blockchain Decentralization - arXiv https://arxiv.org/html/2501.18279v1
18. A Data Engineering Framework for Ethereum Beacon Chain Rewards https://www.nature.com/articles/s41597-025-04623-7

19. Blockchain technology in the energy sector: A systematic review of ... https://www.sciencedirect.com/science/article/pii/S1364032118307184
20. (PDF) Tokenomics in Web3: A Strategic Framework for Sustainable ... https://www.researchgate.net/publication/390936830_Tokenomics_in_Web3_A_Strategic_Framework_for_Sustainable_and_Scalable_Blockchain_Ecosystems
21. Toward the Mass Adoption of Blockchain: Cross-Industry Insights ... https://www.mdpi.com/2504-2289/9/7/178
22. Blockchain in agricultural value chains of developing economies https://www.researchgate.net/publication/391975870_Blockchain_in_agricultural_value_chains_of_developing_economies_-_progress_challenges_and_future_pathways
23. [PDF] Digital Agriculture Roadmap Playbook - World Bank Document https://documents1.worldbank.org/curated/en/099053025063021993/pdf/P508004-f943a09b-c45f-4c93-b554-9dd1decd1e7c.pdf
24. Determinants of Blockchain Technology Adoption in the Australian ... https://www.mdpi.com/2071-1050/16/13/5806
25. Industrial upgrading in China's hog sector - Yuxing Zhang, 2025 https://journals.sagepub.com/doi/10.1177/20539517251367445
26. Blockchain-Powered Traceability in the Wine Industry: Enhancing ... https://www.sciencedirect.com/science/article/pii/S2096720925001320
27. [PDF] A Blockchain Protocol for Human-in-the-Loop AI - arXiv.org https://arxiv.org/pdf/2211.10859
28. Metaversal intelligence: Unifying human-AI interactions in human-in ... https://www.sciencedirect.com/science/article/abs/pii/S1389128625003925
29. A Blockchain Protocol for Human-in-the-Loop AI - ResearchGate https://www.researchgate.net/publication/365634108_A_Blockchain_Protocol_for_Human-in-the-Loop_AI
30. Accidental AIOps Architecture: From Crypto Trading to Network ... https://www.linkedin.com/posts/propatriavigilans_networkautomation-aiops-machinelearning-activity-7431047860827287552-VOZV
31. Fostering AI alignment through blockchain, proof of personhood and ... https://link.springer.com/article/10.1007/s10586-025-05729-8
32. (PDF) Smart Contracts in Agri-Finance Supply Chains - ResearchGate https://www.researchgate.net/publication/396884160_Smart_Contracts_in_Agri-Finance_Supply_Chains
33. Blockchain-Driven Tripartite Game Analysis of Agricultural Supply ... https://www.sciencedirect.com/science/article/pii/S2666188825008147

34. [PDF] Blockchain for agriculture - FAO Knowledge Repository https://openknowledge.fao.org/server/api/core/bitstreams/66ce8120-ed1b-4469-a4ca-08529d3b0774/content
35. [PDF] Smart Contracts for Managing the Agricultural Supply Chain https://univ-reims.hal.science/hal-05224351v1/file/Smart_Contracts_for_Managing_the_Agricultural_Supply_Chain_A_Practical_Case_Study.pdf
36. Blockchain technology for pay-for-outcome sustainable agriculture ... https://iopscience.iop.org/article/10.1088/2515-7620/ad16f0/pdf
37. Blockchain-Based Agri-Food Supply Chain: A Complete Solution https://www.academia.edu/50749430/Blockchain_Based_Agri_Food_Supply_Chain_A_Complete_Solution
38. IEEE Std 2418.5™-2025, IEEE Guide for Blockchain in Power and ... https://ieeexplore.ieee.org/iel8/11301066/11301067/11301068.pdf
39. POLARIS: Cross-Domain Access Control via Verifiable Identity and ... https://arxiv.org/html/2511.22017v1
40. Self sovereign and blockchain based access control https://www.sciencedirect.com/science/article/pii/S1084804522002181
41. SSIBAC in a multi-organizational setting, in light of the XACML... https://www.researchgate.net/figure/SSIBAC-in-a-multi-organizational-setting-in-light-of-the-XACML-standard-perspective_fig1_350453579
42. [PDF] D2.1 - Documentation of available building blocks and ... - HAL https://hal.science/hal-05506413v1/document
43. [PDF] Digital Identity Standards - ENISA https://www.enisa.europa.eu/sites/default/files/publications/Digital_Identity_Standards.pdf
44. Blockchain Reference System Architecture Description for the ISA95 ... https://www.mdpi.com/1424-8220/20/22/6456
45. Verifiable Credentials Data Model v2.0 - W3C https://www.w3.org/TR/vc-data-model-2.0/
46. [PDF] Next Generation Internet of Things - European Commission https://ec.europa.eu/research/participants/documents/downloadPublic?documentIds=080166e5d31993bd&appId=PPGMS
47. (PDF) A Blockchain-Based Tokenization Approach to Empower ... https://www.researchgate.net/publication/401041229_A_Blockchain-Based_Tokenization_Approach_to_Empower_Farmers_and_Youth_in_Smart_Agriculture
48. (PDF) Token Design Strategies for Entrepreneurial Crypto Projects ... https://www.researchgate.net/publication/

397099660_Token_Design_Strategies_for_Entrepreneurial_Crypto_Projects_a_Systemic_Literature_Review

49. IoT and blockchain for trusted sustainable produce - ScienceDirect https://www.sciencedirect.com/science/article/pii/S0168169924005751
50. Blockchain in IoT Networks for Precision Agriculture - ResearchGate https://www.researchgate.net/publication/363777267_Blockchain_in_IoT_Networks_for_Precision_Agriculture
51. IoT and Blockchain-Based Smart Agri-food Supply Chains - Springer https://link.springer.com/rwe/10.1007/978-3-030-69698-6_91
52. (PDF) Blockchain Enabled Architecture with Selective Consensus ... https://www.researchgate.net/publication/366575312_Blockchain_Enabled_Architecture_with_Selective_Consensus_Mechanisms_for_IoT_Based_Saffron-Agri_Value_Chain
53. Blockchain Technology Enabling Applications in Smart Agriculture https://dl.acm.org/doi/fullHtml/10.1145/3696500.3696542
54. Blockchain Repository - Agriculture !! | Prasanna Lohar - LinkedIn https://www.linkedin.com/posts/prasannalohar_agristack-activity-7334103811550031874-cmJE
55. Integrating blockchain technology within an information ecosystem https://www.sciencedirect.com/science/article/pii/S2096720924000381
56. Blockchain Integration: Merging Legacy Systems with Blockchain https://decubate.com/blog/blockchain-integration-merging-legacy-systems-with-blockchain
57. (PDF) Behavioral Analytics and Blockchain: Driving Customer ... https://www.researchgate.net/publication/394292257_Behavioral_Analytics_and_Blockchain_Driving_Customer_Loyalty_and_Corporate_Transparency
58. Digitalizing the Closing-of-the-Loop for Supply Chains - MDPI https://www.mdpi.com/2071-1050/13/5/2895
59. KDD: Knowledge Discovery and Data Mining - ACM Digital Library https://dl.acm.org/doi/proceedings/10.1145/3580305?tocHeading=heading1
60. [PDF] 2026 POMS-HK Book of Abstracts https://sme.cuhk.edu.cn/sites/default/files/2025-12/Book_of_Abstracts_12_25_v1.pdf
61. [PDF] Comprehensive Technical Framework for Blockchain and ... - SEC.gov https://www.sec.gov/files/ctf-written-proposal-sec-project-09-15-2025.pdf
62. Blockchain-Driven Food Supply Chains: A Systematic Review for ... https://www.mdpi.com/2076-3417/14/19/8944
63. Is Blockchain a Panacea for Guarding PDO Supply Chains ... https://advanced.onlinelibrary.wiley.com/doi/full/10.1002/adsu.202400257

64. A strategic analysis of blockchain adoption in collaboration with e ... https://www.researchgate.net/publication/400974170_Empowering_farmers_A_strategic_analysis_of_blockchain_adoption_in_collaboration_with_e-tailers_for_poverty_alleviation
65. Blockchain-Based Frameworks for Food Traceability: A Systematic ... https://www.mdpi.com/2304-8158/12/16/3026
66. A Conceptual UX Model for Designing and Developing the Business ... https://www.researchgate.net/publication/375039638_A_Conceptual_UX_Model_for_Designing_and_Developing_the_Business_Intelligence_Dashboards
67. Accessibility Case Study: Transforming a Complex ... - LinkedIn https://www.linkedin.com/pulse/accessibility-case-study-transforming-complex-he-him--jelnc
68. Decision support system for blockchain (DLT) platform selection ... https://www.sciencedirect.com/science/article/abs/pii/S0957417422017328
69. Consensus in Blockchain Systems with Low Network Throughput https://dl.acm.org/doi/fullHtml/10.1145/3475992.3475995
70. Decision support model for blockchain adoption using EDAS ... https://www.nature.com/articles/s41598-025-05971-5
71. [PDF] Consensus in Blockchain Systems with Low Network Throughput https://arxiv.org/pdf/2103.02916
72. Decision Support for Blockchain Platform Selection: Three Industry ... https://www.researchgate.net/publication/338491432_Decision_Support_for_Blockchain_Platform_Selection_Three_Industry_Case_Studies
73. User Interface of Blockchain-Based Agri-Food Traceability ... https://ieeexplore.ieee.org/iel7/6287639/9312710/09446073.pdf
74. Decision support model for blockchain adoption using EDAS ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC12669690/
75. Blockchain-Based Decentralized Storage Systems for Sustainable ... https://www.mdpi.com/2071-1050/16/17/7671
76. Identifying Agricultural Market Structures and Digital Marketing ... https://dl.acm.org/doi/full/10.1145/3726122.3726172
77. Digital Agriculture for Small-Scale Producers https://cacm.acm.org/research/digital-agriculture-for-small-scale-producers/
78. Smart Agriculture: Current State, Opportunities, and Challenges https://ieeexplore.ieee.org/iel8/6287639/10380310/10701486.pdf
79. An Extensive Blockchain Based Applications Survey - IEEE Xplore https://ieeexplore.ieee.org/iel7/6287639/9668973/09936616.pdf

80. A Comparative Study: Blockchain Technology Utilization Benefits ... http://ieeexplore.ieee.org/iel7/6287639/9312710/09317729.pdf
81. Ontologies for Smart Agriculture: A Path Toward Explainable AI—A ... https://ieeexplore.ieee.org/iel8/6287639/10820123/10973055.pdf
82. Emerging Technologies for Sustainable Agriculture - IEEE Xplore https://ieeexplore.ieee.org/iel8/6287639/10380310/10597387.pdf
83. Understanding Blockchains Through Prototypes and Public ... https://dl.acm.org/doi/full/10.1145/3503462
84. Green AI for Smart Agriculture: Energy-Efficient Predictive Models for ... https://ieeexplore.ieee.org/iel8/6287639/10820123/11267389.pdf
85. AI-powered blockchain technology in industry 4.0, a review https://www.sciencedirect.com/science/article/pii/S2949948824000015
86. Smart Contracts, Blockchain, and Health Policies: Past, Present, and ... https://www.mdpi.com/2078-2489/16/10/853
87. Predictive Analytics and Blockchain for Smart Public Administration https://www.researchgate.net/publication/396087032_AI-Driven_Governance_Predictive_Analytics_and_Blockchain_for_Smart_Public_Administration
88. Blockchain in the Digital Twin Context: A Comprehensive Survey https://dl.acm.org/doi/10.1145/3772366
89. Integration of Blockchain, IoT and Machine Learning for Multistage ... https://pmc.ncbi.nlm.nih.gov/articles/PMC7923442/
90. Survey on Blockchain-Enhanced Machine Learning - IEEE Xplore https://ieeexplore.ieee.org/iel7/6287639/10005208/10366252.pdf
91. Toward Integrated and Trusted Intelligence for Industry 5.0 - arXiv https://arxiv.org/html/2601.04569v1
92. Design and evaluation of a decentralized urban governance system ... https://www.frontiersin.org/journals/sustainable-cities/articles/10.3389/frsc.2025.1623412/full
93. Public service operational efficiency and blockchain – A case study ... https://www.sciencedirect.com/science/article/pii/S0740624X22000958
94. Enhancing EHR Interoperability and Security through Distributed ... https://pmc.ncbi.nlm.nih.gov/articles/PMC11477175/
95. From Data Silos to Health Records Without Borders - MDPI https://www.mdpi.com/2078-2489/16/2/106
96. (PDF) Enhancing Blockchain Interoperability and Intraoperability ... https://www.researchgate.net/publication/

376489036_Enhancing_Blockchain_Interoperability_and_Intraoperability_Capabilities_in_Collaborative_Enterprise-A_Standardized_Architecture_Perspective

97. KEDM Event-Driven Monitor Overview | PDF - Scribd https://www.scribd.com/document/725056018/KEDM-04-14-24-Vol-172
98. United Spirits Limited Annual Report 2023 24 | PDF | Brand - Scribd https://www.scribd.com/document/758070823/United-Spirits-Limited-Annual-Report-2023-24
99. Zero Draft NE Vision Plan 2047 | PDF | Economies - Scribd https://www.scribd.com/document/906059028/Zero-Draft-NE-Vision-Plan-2047
100. An Empirical Exploration of Usability through the PACMAD Model in ... https://dl.acm.org/doi/10.1145/3690001.3690029
101. (PDF) User Interface of Blockchain-Based Agri-Food Traceability ... https://ieeexplore.ieee.org/iel7/6287639/6514899/09446073.pdf
102. An Empirical Exploration of Usability through the PACMAD Model in ... https://dl.acm.org/doi/fullHtml/10.1145/3690001.3690029
103. An Empirical Exploration of Usability through the PACMAD Model in ... https://dl.acm.org/doi/pdf/10.1145/3690001.3690029
104. Convergence of Blockchain, IoT, and AI for Enhanced Traceability ... https://ieeexplore.ieee.org/iel8/6287639/10820123/10836689.pdf
105. Developing and Evaluating a P2P Land-Based Agricultural Market ... https://dl.acm.org/doi/full/10.1145/3691422.3691475
106. Identifying Agricultural Market Structures and Digital ... - ACM https://dl.acm.org/doi/10.1145/3726122.3726172
107. (PDF) An Extensive Blockchain Based Applications Survey https://ieeexplore.ieee.org/iel7/6287639/6514899/09936616.pdf
108. Eghatha: A Blockchain-Based System to Enhance Disaster ... - MDPI https://www.mdpi.com/2073-431X/14/10/405
109. An Overview of Blockchain for Industry 5.0: Towards Human-Centric ... https://ieeexplore.ieee.org/iel8/6287639/10380310/10614164.pdf
110. Blockchain-Based Frameworks for Food Traceability: A Systematic ... https://www.researchgate.net/publication/373079013_Blockchain-Based_Frameworks_for_Food_Traceability_A_Systematic_Review
111. AI revolutionizing industries worldwide: A comprehensive overview ... https://www.sciencedirect.com/science/article/pii/S2773207X24001386
112. [PDF] IMPROVING HUMANITARIAN PAYMENTS THROUGH DIGITAL ... https://btca-production-site.s3.amazonaws.com/documents/616/english_attachments/

Improving_Humanitarian_Payments_Through_Digital_Innovation_%E2%80%93_Challenges_and_Opportunities.pdf

113. [PDF] An Explainable Decision Support System for Predictive Process ... https://arxiv.org/pdf/2207.12782
114. (PDF) Real-Time Analytics Dashboards for Decision-Making Using ... https://www.researchgate.net/publication/393907578_Real-Time_Analytics_Dashboards_for_Decision-Making_Using_Tableau_in_Public_Sector_and_Business_Intelligence_Applications
115. Learning Analytics: The Ultimate Guide | DLI Blog https://www.digitallearninginstitute.com/blog/learning-analytics-the-ultimate-guide
116. Understanding the Differences Between Descriptive, Predictive, and ... https://www.coursera.org/articles/descriptive-predictive-and-prescriptive-analytics
117. The evaluation of a decision support system integrating assistive ... https://pmc.ncbi.nlm.nih.gov/articles/PMC11285597/
118. Blockchain in Communication Networks: A Comprehensive Review https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/blc2.70031
119. Blockchain-Based Cloud Manufacturing SCM System for ... - MDPI https://www.mdpi.com/2076-3417/12/17/8664
120. Blockchain Interoperability Issues in Supply Chain - Academia.edu https://www.academia.edu/115226732/Blockchain_Interoperability_Issues_in_Supply_Chain_Exploration_of_Mass_Adoption_Procedures
121. (PDF) Blockchain Technology in Agriculture: A Case Study of ... https://www.researchgate.net/publication/355705080_Blockchain_Technology_in_Agriculture_A_Case_Study_of_Blockchain_Start-Up_Companies
122. A Systematic Literature Review of Blockchain-Enabled Supply ... https://www.mdpi.com/2071-1050/14/4/2439
123. A global blockchain-based agro-food value chain to facilitate trade ... https://www.nature.com/articles/s41599-023-01658-2
124. Blockchain technology for agricultural supply chains during ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC8908783/
125. [PDF] Artificial Intelligence for Agriculture Innovation https://www3.weforum.org/docs/WEF_Artificial_Intelligence_for_Agriculture_Innovation_2021.pdf
126. Development of a Predictive Dashboard With Prescriptive Decision ... https://pubmed.ncbi.nlm.nih.gov/40193194/
127. Design principles for learning analytics information systems in ... https://www.tandfonline.com/doi/full/10.1080/0960085X.2020.1816144

128. Orchestrating Blockchain for Sustainability-oriented Supply chains https://www.sciencedirect.com/science/article/pii/S0925527326000587?dgcid=rss_sd_all
129. [PDF] Farm data management, sharing and services for agriculture ... https://openknowledge.fao.org/server/api/core/bitstreams/7043933b-dd27-4674-92ec-4bbfef8a4d48/content
130. [PDF] Economic aspects of national telecommunications/ICTs - ITU https://www.itu.int/dms_pub/itu-d/opb/stg/D-STG-SG01.04.1-2025-PDF-E.pdf
131. [2602.09608] Designing a Token Economy: Incentives, Governance ... https://arxiv.org/abs/2602.09608
132. Designing a Token Economy: Incentives, Governance, and ... https://arxiv.org/html/2602.09608v1
133. The Rise of Blockchain Technology in Agriculture and Food Supply ... https://www.researchgate.net/publication/334805119_The_Rise_of_Blockchain_Technology_in_Agriculture_and_Food_Supply_Chains
134. (PDF) Blockchain technology in agri-food value chain management https://www.researchgate.net/publication/332903007_Blockchain_technology_in_agri-food_value_chain_management_A_synthesis_of_applications_challenges_and_future_research_directions
135. Blockchain technology in current agricultural systems - ResearchGate https://www.researchgate.net/publication/343375975_Blockchain_technology_in_current_agricultural_systems_from_techniques_to_applications
136. A Systematic Literature Review on Applications of Information and ... https://www.researchgate.net/publication/350333130_A_Systematic_Literature_Review_on_Applications_of_Information_and_Communication_Technologies_and_Blockchain_Technologies_for_Precision_Agriculture_Development
137. Application of digital technologies for ensuring agricultural productivity https://www.researchgate.net/publication/375810263_Application_of_digital_technologies_for_ensuring_agricultural_productivity
138. (PDF) A systematic literature review of blockchain-based applications https://www.researchgate.net/publication/329136952_A_systematic_literature_review_of_blockchain-based_applications_Current_status_classification_and_open_issues
139. (PDF) Digital Innovation, Business Models Transformations, and ... https://www.researchgate.net/publication/

394618688_Digital_Innovation_Business_Models_Transformations_and_Agricultural_SMEs_A_PRISMA-Based_Review_of_Challenges_and_Prospects

140. (PDF) An overview of food supply chain virtualization and granular ... https://www.researchgate.net/publication/338447663_An_overview_of_food_supply_chain_virtualization_and_granular_traceability_using_blockchain_technology
141. Blockchain Integration in the Era of Industrial Metaverse https://www.researchgate.net/publication/367324333_Blockchain_Integration_in_the_Era_of_Industrial_Metaverse
142. (PDF) Enhancing adaptive and sustainable resilience through ... https://www.researchgate.net/publication/400992771_Enhancing_adaptive_and_sustainable_resilience_through_artificial_intelligence_machine_learning_internet_of_things_big_data_analytics_and_blockchain
143. (PDF) Manual of ICT Tools in Agricultural Extension - ResearchGate https://www.researchgate.net/publication/397572021_Manual_of_ICT_Tools_in_Agricultural_Extension
144. Learning analytics dashboard: a tool for providing actionable ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC8853217/
145. Use of Predictive Analytics within Learning Analytics Dashboards https://www.researchgate.net/publication/362975782_Use_of_Predictive_Analytics_within_Learning_Analytics_Dashboards_A_Review_of_Case_Studies
146. Data-Driven Prescriptive Analytics Applications: A Comprehensive ... https://arxiv.org/html/2412.00034v3
147. Use of Predictive Analytics within Learning Analytics Dashboards https://link.springer.com/article/10.1007/s10758-022-09613-x
148. Heuristics for Supporting Cooperative Dashboard Design https://ieeexplore.ieee.org/iel7/2945/10373160/10297573.pdf
149. (PDF) Predictive and Prescriptive Analytics in Business Decision ... https://www.researchgate.net/publication/366656964_Predictive_and_Prescriptive_Analytics_in_Business_Decision_Making_Needs_and_Concerns
150. Contemporary Business Analytics: An Overview - MDPI https://www.mdpi.com/2306-5729/6/8/86
151. [PDF] Predictive and Prescriptive Analytics in Business Decision Making https://arxiv.org/pdf/2212.13643
152. A Blockchain-Enabled Architecture for Secure and Transparent ... https://www.mdpi.com/2079-8954/14/2/171

153. Artificial intelligence in humanitarian aid: A review and future ... https://www.sciencedirect.com/science/article/pii/S0166497225002470

154. An Empirical Study on the Application of Blockchain Technology in E ... https://www.researchgate.net/publication/372383988_An_Empirical_Study_on_the_Application_of_Blockchain_Technology_in_E-Agriculture_An_Innovative_Service_UIS_Application_System

155. Blockchain-based Data Provenance for the Internet of Things https://www.researchgate.net/publication/335172596_Blockchain-based_Data_Provenance_for_the_Internet_of_Things

156. [PDF] Towards a data provenance model for private data sharing ... https://univ-pau.hal.science/hal-03468281/document

157. Internet of Things and Blockchain Adoption in Food Supply Chain https://www.mdpi.com/2624-831X/6/3/51

158. A blockchain- and self-sovereign identity-based collaborative ... https://www.sciencedirect.com/science/article/pii/S2772375525008858

159. (PDF) Internet of Things (IoT) in Supply Chain Management https://www.researchgate.net/publication/374560424_Internet_of_Things_IoT_in_Supply_Chain_Management_Challenges_Opportunities_and_Best_Practices

160. Implementation of blockchain technology in integrated IoT networks ... https://www.sciencedirect.com/science/article/pii/S2096720924000010

161. Digitalization and Blockchain Integration in Agri-Food Supply Chains https://www.mdpi.com/2071-1050/17/20/9276

162. Monetize Farm Data with Blockchain: Who Pays You & Why https://gardenagrotech.com/monetize-farm-data-with-blockchain/

163. Leveraging Advanced Technologies: Business Model Innovation ... https://link.springer.com/content/pdf/10.1007/978-3-032-00447-5.pdf

164. (PDF) Designing a Token Economy: Incentives, Governance and ... https://www.researchgate.net/publication/371314053_Designing_a_Token_Economy_Incentives_Governance_and_Tokenomics

165. (PDF) Beyond The Hype: a Real World Evaluation of Blockchain's ... https://www.researchgate.net/publication/395394731_BEYOND_THE_HYPE_A_REAL_WORLD_EVALUATION_OF_BLOCKCHAIN'S_ROLE_IN_DEMOCRATIZING_AGRI_FOOD_SYSTEMS

166. Leveraging Computer Technology for Digital Village Development https://www.sciencedirect.com/org/science/article/pii/S1947319226000079

167. (PDF) A Systematic Literature Review of Blockchain Technology for ... https://www.researchgate.net/publication/

355683738_A_Systematic_Literature_Review_of_Blockchain_Technology_for_Smart_Villages

168. Leveraging Computer Technology for Digital Village Development https://www.researchgate.net/publication/399833154_Leveraging_Computer_Technology_for_Digital_Village_Development_Theoretical_Insights_and_Practical_Approaches
169. (PDF) Feasibility of Low-Code Development Platforms in Precision ... https://www.researchgate.net/publication/384368411_Feasibility_of_Low-Code_Development_Platforms_in_Precision_Agriculture_Opportunities_Challenges_and_Future_Directions
170. Empirical research on the blockchain adoption – based on TAM https://www.researchgate.net/publication/350385017_Empirical_research_on_the_blockchain_adoption_-_based_on_TAM
171. A mixed method approach to integrate digitization and sustainability ... https://www.sciencedirect.com/science/article/pii/S277237552500245X
172. On the Integration of Artificial Intelligence and Blockchain Technology https://www.researchgate.net/publication/377073122_On_the_Integration_of_Artificial_Intelligence_and_Blockchain_Technology_a_Perspective_about_Security
173. A beginner's guide to infrastructure − less networking concepts https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ntw2.12094
174. Algorithm aversion in agricultural decision-making: Trust dynamics ... https://www.sciencedirect.com/science/article/pii/S0308521X25003701
175. (PDF) Integrated Decision Support Systems (IDSS) for Dairy Farming https://www.researchgate.net/publication/353047687_Integrated_Decision_Support_Systems_IDSS_for_Dairy_Farming_A_Discussion_on_How_to_Improve_Their_Sustained_Adoption
176. Scaling Decision − Support Tools to Promote Soil Health: Insights ... https://bsssjournals.onlinelibrary.wiley.com/doi/full/10.1111/ejss.70113
177. Resource-Governed BDA Adoption for Resilient Supply-Chain ... https://www.mdpi.com/2071-1050/17/21/9620
178. Mathematics for action: supporting science-based decision-making https://unesdoc.unesco.org/ark:/48223/pf0000380883.locale=en
179. (PDF) Smart contracts and automation in supply chain financing https://www.researchgate.net/publication/398909729_Smart_contracts_and_automation_in_supply_chain_financing
180. Prescriptive analytics motivating distance learning students to take ... https://dl.acm.org/doi/full/10.1145/3706468.3706508

181. (PDF) Blockchain in Agriculture Traceability Systems: A Review https://www.researchgate.net/publication/342188159_Blockchain_in_Agriculture_Traceability_Systems_A_Review
182. Survey on the Applications of Blockchain in Agriculture - MDPI https://www.mdpi.com/2077-0472/12/9/1333
183. Design of a low-cost smart irrigation system based on NB-IOT ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC12758771/
184. [PDF] Intelligent Economy - Huawei https://www.huawei.com/-/media/CORPORATE/minisite/giv2030/pdf/intelligent-economy-final-en.pdf
185. A Blockchain and Artificial Intelligence-Based, Patient-Centric ... - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC8391524/
186. (PDF) Using Blockchain to Track AI Model Origins - ResearchGate https://www.researchgate.net/publication/395414943_Using_Blockchain_to_Track_AI_Model_Origins
187. [PDF] Blockchain-based digital twin data provenance for predictive asset ... https://pdfs.semanticscholar.org/9354/69e69ac250d25980f6a905f881bf8183ab27.pdf
188. Securing AI − based healthcare systems using blockchain technology ... https://onlinelibrary.wiley.com/doi/10.1002/ett.4884
189. ai-enabled decision support systems for smarter infrastructure ... https://www.researchgate.net/publication/393655215_AI-ENABLED_DECISION_SUPPORT_SYSTEMS_FOR_SMARTER_INFRASTRUCTURE_PROJECT_MANAGEMENT_IN_PUBLIC_WORKS
190. [PDF] World Energy Markets OBSERVATORY 2023 - Capgemini https://www.capgemini.com/wp-content/uploads/2023/11/Capgemini_WEMO_2023-report.pdf
191. (PDF) INFORMATION GOVERNANCE - Academia.edu https://www.academia.edu/43777886/INFORMATION_GOVERNANCE
192. Chief Transformation Transformation Officer Hand | PDF - Scribd https://www.scribd.com/document/912691707/Chief-Transformation-Transformation-Officer-Hand
193. AIoT-Enabled Data Management for Smart Agriculture - ResearchGate https://www.researchgate.net/publication/392605537_AIoT-Enabled_Data_Management_for_Smart_Agriculture_A_Comprehensive_Review_on_Emerging_Technologies
194. Section 1: Careers and Roles Flashcards | Quizlet https://quizlet.com/840664317/section-1-careers-and-roles-flash-cards/
195. Data Science and Analytics: An Overview from Data-Driven Smart ... https://pmc.ncbi.nlm.nih.gov/articles/PMC8274472/

196. Descriptive, predictive, diagnostic, and prescriptive analytics explained https://business.adobe.com/blog/basics/descriptive-predictive-prescriptive-analytics-explained

197. [Solved] TOPIC: Descriptive, Prescriptive and Predictive Analytics ... https://www.coursehero.com/tutors-problems/Management/39737138-TOPIC-Descriptive-Prescriptive-and-Predictive-Analytics-/

198. Integration of Privacy Protection and Blockchain-Based Food Safety ... https://pmc.ncbi.nlm.nih.gov/articles/PMC9367899/

199. (PDF) Strategic considerations for digitalising humanitarian supply ... https://www.researchgate.net/publication/400257264_Strategic_considerations_for_digitalising_humanitarian_supply_chains_for_resilience_A_bibliometric_analysis

200. Past Groups - W3C https://www.w3.org/community/groups/past/

201. [PDF] AU Digital Agriculture Strategy (DAS) and Implementation Plan 2024 https://faolex.fao.org/docs/pdf/au225158.pdf

202. [PDF] STATUS OF DIGITAL AGRICULTURE IN 47 SUB-SAHARAN ... - ITU https://www.itu.int/dms_pub/itu-d/opb/str/D-STR-DIG_AFR.01-01-2022-PDF-E.pdf

203. Lifecycle management of urban renewal enabled by Internet of Things https://www.sciencedirect.com/science/article/pii/S2590123025017773

204. [PDF] Accelerating the SDGs through digital public infrastructure https://www.undp.org/sites/g/files/zskgke326/files/2023-12/undp-accelerating-the-sdgs-through-digital-public-infrastructure-v2.pdf

205. An IoT-based smart emotion recognition system by using internal ... https://pmc.ncbi.nlm.nih.gov/articles/PMC12923801/

206. Research on the Application of Data Intelligence-Based Intelligent ... https://dl.acm.org/doi/10.1145/3773365.3773623

207. Blockchain-integrated AI framework for secure IoT-based digital ... https://link.springer.com/article/10.1007/s43926-025-00262-x

208. (PDF) Generative AI-Driven Smart Contract Optimization for Secure ... https://www.researchgate.net/publication/393757474_Generative_AI-Driven_Smart_Contract_Optimization_for_Secure_and_Scalable_Smart_City_Services

209. A Blockchain-Assisted Federated Learning Framework for Secure ... https://www.mdpi.com/1999-5903/17/1/13

210. A Deep Dive into AI Ethics and Regulations - arXiv https://arxiv.org/html/2412.04683v2

211. (PDF) Unlocking Blockchain's Potential in Supply Chain Management https://www.researchgate.net/publication/

394993636_Unlocking_Blockchain's_Potential_in_Supply_Chain_Management_A_Review_of_Challenges_Applications_and_Emerging_Solutions

212. Multi-Sensor Monitoring, Intelligent Control, and Data Processing for ... https://www.mdpi.com/1424-8220/25/19/6134

213. (PDF) Automated Supply Chain Management in Agriculture Using ... https://www.researchgate.net/publication/389490297_Automated_Supply_Chain_Management_in_Agriculture_Using_Blockchain_and_IoT

214. A Systematic Review of State-of-the-Art TinyML Applications in ... https://ieeexplore.ieee.org/iel8/6287639/10820123/11266886.pdf

215. A SYSTEMATIC LITERATURE REVIEW OF USER-CENTRIC ... https://www.researchgate.net/publication/393730722_A_SYSTEMATIC_LITERATURE_REVIEW_OF_USER-CENTRIC_DESIGN_IN_DIGITAL_BUSINESS_SYSTEMS_ENHANCING_ACCESSIBILITY_ADOPTION_AND_ORGANIZATIONAL_IMPACT

216. SIGCSE: Computer Science Education - ACM https://dl.acm.org/doi/proceedings/10.1145/3641554?tocHeading=heading2

217. STREAM: A Semantic Transformation and Real-Time Educational ... https://www.mdpi.com/1999-5903/17/12/564

218. AI & ML-Frontiers in Cross Disciplinary Applications & Case Studies https://link.springer.com/content/pdf/10.1007/978-3-032-17300-3.pdf

219. [PDF] User Behaviour Analysis and IoT Secure Applications in Smart City ... https://core.ac.uk/download/301580068.pdf

220. (PDF) Beyond Performativity: Inclusive Marketing as A Tool of Social ... https://www.researchgate.net/publication/395290700_Beyond_Performativity_Inclusive_Marketing_as_a_Tool_of_Social_Transformation

221. [PDF] national conference on engineering innovations in emerging ... - ijsrst https://ijsrst.com/paper/v12i15.pdf

222. Computational Science and Its Applications – ICCSA 2025 https://link.springer.com/content/pdf/10.1007/978-3-031-96962-1.pdf

223. Computers, Volume 15, Issue 1 (January 2026) – 70 articles - MDPI https://www.mdpi.com/2073-431X/15/1

224. [DOC] T22-TSAG-221212-TD-GEN-0020!!MSW-A.docx - ITU https://www.itu.int/dms_pub/itu-t/md/22/tsag/td/221212/GEN/T22-TSAG-221212-TD-GEN-0020!!MSW-A.docx

225. Computer Science - arXiv.org https://arxiv.org/list/cs/new

226. A comprehensive survey on securing the social internet of things https://pmc.ncbi.nlm.nih.gov/articles/PMC12624021/